

Linux module 3 – Assignment

Que: Learn about different filesystem types (e.g., ext4, NTFS).

Ans: FAT (File Allocation Table)

- Types: FAT32

- Description: Simple, older file system used in small storage devices like USB drives

and memory cards.

- Strengths: Broad compatibility, simple structure.

- Weaknesses: Limited file size (max 4GB for FAT32).

-. NTFS (New Technology File System)

- Description: Default for Windows OS with advanced features like file permissions and

journaling.

- Strengths: Large file support, security features, data integrity.

- Weaknesses: Limited support outside Windows.

Ext (Extended File System)

- Types: Ext2, Ext3, Ext4

- Description: Default file system for Linux, with Ext4 being the most advanced.

- Strengths: Stable, supports large files and journaling.

- Weaknesses: Not natively supported by Windows.

Que: Manage disk partitions and filesystems using tools like fdisk, mkfs, and mount.

Ans: fdisk: A tool used to create, delete, or modify disk partitions on a system. It works

with MBR (Master Boot Record) or GPT (GUID Partition Table) partition schemes.

Common fdisk commands:

- fdisk -l - List all partitions.
- fdisk /dev/sdX - Start the fdisk utility on a specified disk
- Inside fdisk:
 - m for help
 - p for print partition table
 - n to create a new partition
 - d to delete a partition
 - w to write changes
- mkfs: A tool used to create a filesystem on a partition. You can specify the filesystem type (e.g., ext4, xfs).

Example:

- mkfs.ext4 /dev/sdX1 – Create an ext4 filesystem on the first partition of a disk.
- mkfs.xfs /dev/sdX1 – Create an XFS filesystem.
- mount: This command is used to mount a filesystem to a directory.

Example:

- mount /dev/sdX1 /mnt – Mount the partition /dev/sdX1 to /mnt.

Que: create a 2048MB partition and verify if the partition has been created.

Ans: Done in Lab

Que. Why LVM is required?

Ans: LVM (Logical Volume Manager) is a system for managing disk drives and partitions in

a more flexible and scalable manner compared to traditional partitioning methods.

Que. How can you find out how much memory Linux is using?

Ans: To show how much memory using linux - **df -h**

Que. What is a typical size for a swap partition under a Linux system?

Ans: The typical size for a swap partition depends on the amount of physical RAM and the

intended use of the system.

Que. What is the maximum file size on the ext4 file system?

Ans: The maximum file size on an ext4 filesystem is 16 TiB (Tebibytes).

Que. What is the maximum file size on the xfs file system

Ans: The maximum file size on an XFS filesystem is 8 Exabytes (EB) (or 8 billion gigabytes). This is far larger than the maximum file size on ext4 and is more than

sufficient for typical storage needs